

Homework 2

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1. (40 points) Consider $x_{k+1} = Ax_k + B_k\sigma(C_kx_k + b_k)$, where σ is the ReLU activation function. We know that if there exist a positive definite diagonal matrix Λ_k such that the following matrix inequality holds

$$\begin{bmatrix} A_k^\top A_k - I & A_k^\top B_k \\ B_k^\top A_k & B_k^\top B_k \end{bmatrix} + \begin{bmatrix} C_k & 0 \\ 0 & I \end{bmatrix}^\top \begin{bmatrix} 0 & \Lambda_k \\ \Lambda_k & -2\Lambda_k \end{bmatrix} \begin{bmatrix} C_k & 0 \\ 0 & I \end{bmatrix} \leq 0, \quad (1)$$

then we have $\|x'_{k+1} - x_{k+1}\| \leq \|x'_k - x_k\|$. Apply the above condition to show

- (a) The convex potential layer $x_{k+1} = x_k - \frac{2}{\|W_k\|^2} W_k \sigma(W_k^\top x + b_k)$ is 1-Lipschitz.
- (b) The layer $x_{k+1} = x_k - 2W_k \sigma(W_k^\top x_k + b_k)$ with $W_k^\top W_k = I$ is 1-Lipschitz.
- (c) The layer $x_{k+1} = x_k - 2W_k \text{diag}(\sum_{j=1}^n |W_k^\top W_k|_{ij})^{-1} \sigma(W_k^\top x_k + b_k)$ is 1-Lipschitz (here we assume $\sum_{j=1}^n |W_k^\top W_k|_{ij} \neq 0$ for all i , and n is the column dimension of $W_k^\top W_k$).
- (d) The layer $x_{k+1} = \sqrt{2}M_k^\top \Psi_k \sigma(\sqrt{2}\Psi_k^{-1}N_k x_k + b_k)$ with Ψ_k being any positive definite diagonal matrix and (M_k, N_k) being any matrix pair satisfying $M_k M_k^\top + N_k N_k^\top = I$ is 1-Lipschitz.

2. (10 points) Consider the deep equilibrium model $z = \sigma(Wz + Ux + b_z)$ with σ being ReLU. Suppose the model is set up in a way that there is a unique solution z for any input x (so we assume the deep equilibrium model is well-posed). Derive a matrix inequality condition to ensure $\|z' - z\| \leq L\|x' - x\|$ for any (x', x) . (Notice that z is the output of x , and z' is the output of x' .)

Hint: Since σ is ReLU, we must have

$$\begin{bmatrix} y' - y \\ \sigma(y') - \sigma(y) \end{bmatrix}^\top \begin{bmatrix} 0 & \Lambda_k \\ \Lambda_k & -2\Lambda_k \end{bmatrix} \begin{bmatrix} y' - y \\ \sigma(y') - \sigma(y) \end{bmatrix} \geq 0$$

where Λ_k is any positive definite diagonal matrix. Use the above quadratic constraint to derive the matrix inequality condition.

3. (20 points) In this problem, you will be asked to test a few LMIs for stochastic gradient descent (SGD). In the class, we talked about how to analyze SGD under the assumptions that f is m -strongly convex and f_i is L -smooth and convex. As in Lecture 11, $f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x)$, and the sample indices i_k are independent and uniform on $\{1, \dots, n\}$. Here you are asked to analyze SGD under a different set of assumptions:

- f satisfies the “one-point convexity” condition: f has a unique global minimizer x^* and for all x , one has $(x - x^*)^\top \nabla f(x) \geq m\|x - x^*\|^2$ with $m > 0$.
- f_i is L -smooth for all i .

Under the above assumptions, f and f_i are not convex in general. However, you can still obtain a convergence bound for SGD under these assumptions.

(a) Suppose $v_k = x_k$ and $w_k = \nabla f_{i_k}(v_k)$. Based on the above assumptions, the following two inequalities hold

$$\begin{aligned} \mathbb{E} \begin{bmatrix} v_k - x^* \\ w_k \end{bmatrix}^\top \begin{bmatrix} -2L^2I & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} v_k - x^* \\ w_k \end{bmatrix} &\leq \frac{2}{n} \sum_{i=1}^n \|\nabla f_i(x^*)\|^2 = M \\ \mathbb{E} \begin{bmatrix} v_k - x^* \\ w_k \end{bmatrix}^\top \begin{bmatrix} 2mI & -I \\ -I & 0 \end{bmatrix} \begin{bmatrix} v_k - x^* \\ w_k \end{bmatrix} &\leq 0 \end{aligned}$$

Use the above inequalities to prove that for $0 \leq \rho < 1$, if there exist non-negative λ_1 and λ_2 such that

$$\begin{bmatrix} 1 - \rho^2 & -\alpha \\ -\alpha & \alpha^2 \end{bmatrix} - \lambda_1 \begin{bmatrix} -2L^2 & 0 \\ 0 & 1 \end{bmatrix} - \lambda_2 \begin{bmatrix} 2m & -1 \\ -1 & 0 \end{bmatrix} \leq 0, \quad (2)$$

then SGD satisfies the bound

$$\mathbb{E}\|x_k - x^*\|^2 \leq \rho^{2k} \mathbb{E}\|x_0 - x^*\|^2 + \frac{\lambda_1 M}{1 - \rho^2}$$

(b) For $0 < \alpha < m/L^2$, use the above LMI condition to show that SGD satisfies the following bound:

$$\mathbb{E}\|x_k - x^*\|^2 \leq (1 - 2m\alpha + 2L^2\alpha^2)^k \mathbb{E}\|x_0 - x^*\|^2 + \frac{\alpha^2 M}{2m\alpha - 2L^2\alpha^2}$$

4. (30 points) In this problem, you will be asked to perform several calculations, and these calculations eventually lead to the convergence rate proof for Nesterov's accelerated method applied to smooth strongly-convex objective functions. Recall Nesterov's method is defined by the following recursion:

$$x_{k+1} = x_k + \beta(x_k - x_{k-1}) - \alpha \nabla f((1 + \beta)x_k - \beta x_{k-1})$$

which can also be written as

$$\xi_{k+1} = A\xi_k + Bw_k$$

$$v_k = C\xi_k$$

$$w_k = \nabla f(v_k)$$

where $A = \begin{bmatrix} (1 + \beta)I & -\beta I \\ I & 0 \end{bmatrix}$, $B = \begin{bmatrix} -\alpha I \\ 0 \end{bmatrix}$, $C = [(1 + \beta)I \quad -\beta I]$, and $\xi_k = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$.

(a) Assume f is L -smooth and m -strongly convex. By L -smoothness and m -strong convexity, we have

$$\begin{aligned} f(x_k) - f(x_{k+1}) &= f(x_k) - f(v_k) + f(v_k) - f(x_{k+1}) \\ &\geq \nabla f(v_k)^\top (x_k - v_k) + \frac{m}{2} \|x_k - v_k\|^2 \\ &\quad + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2} \|v_k - x_{k+1}\|^2 \\ &= \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix}^\top X_1 \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix} \end{aligned}$$

The last step in the above derivation requires substituting $x_{k+1} = (1 + \beta)x_k - \beta x_{k-1} - \alpha \nabla f(v_k)$ and $v_k = C\xi_k$ into the second-to-last line $\nabla f(v_k)^\top (x_k - v_k) + \frac{m}{2} \|x_k - v_k\|^2 + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2} \|v_k - x_{k+1}\|^2$ and rewriting the resultant quadratic function. Your task is figuring out this symmetric matrix X_1 (actually this matrix has already been given in the lecture note).

(b) Similarly, by L -smoothness and m -strong convexity, we have

$$\begin{aligned} f(x^*) - f(x_{k+1}) &= f(x^*) - f(v_k) + f(v_k) - f(x_{k+1}) \\ &\geq \nabla f(v_k)^\top (x^* - v_k) + \frac{m}{2} \|x^* - v_k\|^2 \\ &\quad + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2} \|v_k - x_{k+1}\|^2 \\ &= \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix}^\top X_2 \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix} \end{aligned}$$

The last step in the above derivation requires substituting $x_{k+1} = (1+\beta)x_k - \beta x_{k-1} - \alpha \nabla f(v_k)$ and $v_k = C\xi_k$ into the second-to-last line $\nabla f(v_k)^\top (x^* - v_k) + \frac{m}{2}\|x^* - v_k\|^2 + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2}\|v_k - x_{k+1}\|^2$ and rewriting the resultant quadratic function. Your task is figuring out this symmetric matrix X_2 .

(c) Now based on the inequalities in (a) and (b), you can simply choose $X = \rho^2 X_1 + (1 - \rho^2)X_2$ for any $0 \leq \rho < 1$, and we have

$$\begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix}^\top X \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix} \leq -(f(x_{k+1}) - f(x^*)) + \rho^2(f(x_k) - f(x^*))$$

Based on the testing condition presented in the class, if there exists $P \geq 0$ such that

$$\begin{bmatrix} A^\top P A - \rho^2 P & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix} - X \leq 0 \quad (3)$$

then the following inequality holds

$$\begin{aligned} & (\xi_{k+1} - \xi^*)^\top P (\xi_{k+1} - \xi^*) - \rho^2 (\xi_k - \xi^*)^\top P (\xi_k - \xi^*) \\ & \leq \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix}^\top X \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix} \\ & \leq -(f(x_{k+1}) - f(x^*)) + \rho^2(f(x_k) - f(x^*)) \end{aligned}$$

which directly leads to the linear convergence rate for Nesterov's method:

$$(\xi_{k+1} - \xi^*)^\top P (\xi_{k+1} - \xi^*) + f(x_{k+1}) - f(x^*) \leq \rho^2 ((\xi_k - \xi^*)^\top P (\xi_k - \xi^*) + f(x_k) - f(x^*)).$$

Finding P to satisfy (3) is not trivial. Your task is to verify (3) holds with $P = \frac{1}{2} \begin{bmatrix} \sqrt{L}I \\ (\sqrt{m} - \sqrt{L})I \end{bmatrix} \begin{bmatrix} \sqrt{L}I & (\sqrt{m} - \sqrt{L})I \end{bmatrix} \geq 0$, $\rho^2 = 1 - \sqrt{\frac{m}{L}}$, $\alpha = \frac{1}{L}$, and $\beta = \frac{\sqrt{L} - \sqrt{m}}{\sqrt{L} + \sqrt{m}}$. This gives a complete proof for the accelerated rate of Nesterov's method.

(Hint: The calculation here can be lengthy. So you are allowed to use some symbolic toolbox to help as long as you turn in the code.)